

| Question | Working | Answer | Mark               | Notes                                                                                                                                                                                                                                                                                                                                                                                  |
|----------|---------|--------|--------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 16       |         |        | M1<br>M1<br><br>C1 | for the start of a method to convert 0.22.. to a fraction, eg $10y = 2.22..$ or $(y=) \frac{2}{9}$<br>for the start of a method to convert 0.13636... to a fraction,<br>$10x = 1.3636..$ or $100x = 13.6363...$ or $1000x = 136.3636..$ or $(x=) \frac{13.5}{99}$ or $(x=) \frac{135}{990}$<br><br>for correct arithmetic and concluding the proof<br><br>OR<br><br>M1<br>M1<br><br>C1 |
|          |         |        |                    | for $0.1\dot{3}\dot{6} \times 0.\dot{2} = 0.\dot{0}\dot{3} (= z)$<br>for complete method to find two appropriate recurring decimals the difference of which is<br>a rational number, eg. $100z = 3.0303..., (z =) 0.0303... \text{ or } \frac{3}{99}$<br>for correct arithmetic and concluding the proof                                                                               |